PROJECT PROPOSAL

Book Mood Reader

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Introduction

1. What?

The Book Mood Reader is a project that creates music playlists based on the mood and atmosphere of a book. It is made for readers who enjoy listening to music while reading and want a playlist that fits the story they are reading.

The project gives the reader a longer selection of songs that match the feeling of the book, instead of relying only on the short playlists that are sometimes included at the beginning of a book. The goal is to make the reading experience more enjoyable and keep the music connected to the story for a longer time.

2. Why?

Some people enjoy listening to music while reading, including me. Some books already have a short playlist at the beginning, but most of the time the songs are not enough for the whole time you spend reading. After a while, the same songs start repeating.

Because I often have this issue myself, I decided to use it to formulate my project. I want to create a solution that can give the reader a longer playlist that still fits the mood of the book and makes the reading experience more enjoyable.

3. Who?

The main people involved in this project will be the users, an expert and my teachers.

- **Primary users** – readers who enjoy listening to music while reading. They will be involved in the creation of the dataset by helping with the annotation of books and selecting moods that they feel describe them. Later, they can also test the final result and give feedback on whether the suggested music actually fits the feeling of the book.
- **Domain expert** – someone with strong knowledge of books and music. The expert can help me understand how the atmosphere of a book can be connected to music and give feedback on the mood labels I use. Their opinion can also help me check whether the matches between books and songs make sense.
- **Teachers** – project and technical guidance. My teachers will guide me during the project by giving advice, feedback and helping me make decisions about the modelling, data and technical side of the project. Their feedback will also help me improve the project during the different stages.
- **Peers or classmates** – additional feedback and testing. They can help me by giving another point of view on the idea, testing parts of the project and giving feedback when something is unclear or needs improvement.

4. When?

The project will be developed over 9 weeks following the AMD methodology:

Week	Planning
Week 1	Ideation and defining the project direction, problem and goal. Start researching the domain and similar solutions.
Week 2	Project planning and Domain Understanding. Define the stakeholders, research questions and what is needed for the project.
Week 3	**Iteration 0** – define the analytical approach, data requirements, mood labels and plan how the data will be collected and annotated.
Week 4	Start collecting and annotating the book data. Prepare and clean the first version of the dataset.
Week 5	Start experimenting with different modelling approaches and evaluate the first results.
Week 6	Continue improving the model and start working on the music data and mood matching.
Week 7	Connect the different parts of the project and create the first complete working version.
Vacation week	Extra time that can be used for unfinished work, further experiments, improvements or additional data collection.
Week 8	Test the project with users and the expert, collect feedback and improve the results based on the testing.
Week 9	Final improvements, evaluation, documentation, reflection and preparation for the final presentation.

5. How?

I will develop the project step by step, starting with the data and the mood labels that will be used for the books. After collecting and preparing the data, I will test a few different modelling approaches and compare the results to see which one works best for predicting the mood of a book.

Once the book mood prediction is working well enough, I will continue with the music side of the project. The predicted moods will be used to find songs that fit the same feeling, and these songs will then be shown to the user as a playlist.
The final product will be a simple working prototype where the user can choose or enter a book and receive a playlist that matches its mood and atmosphere. For the inferencing, the model will take the description of a new book, predict the moods connected to it and use those moods to find suitable songs.
During the final delivery, I will demonstrate how the whole process works, show the prototype, explain the choices I made during the project and present the results from the testing and feedback.

Domain Understanding

The main domain of this project is the connection between **reading, music and emotion**. Both books and music can create certain feelings on their own, but when they are experienced together, the music can influence how the atmosphere of the story is felt by the reader.

The mood and atmosphere of a book

When reading a book, the experience is not created only by what is happening in the story. The setting, characters, relationships, writing style and events can all contribute to a certain atmosphere. A romance can feel warm, emotional or melancholic, while a thriller can create tension, fear or mystery. At the same time, one book usually cannot be described with only one mood. A story can be romantic but also sad, mysterious and intense at different moments.

This is important for my project because I do not want to put a book into only one simple category such as "happy" or "sad". I want to look at the combination of moods that describe the overall feeling of the book. This is also why using multiple mood labels makes more sense for the project. A more detailed mood description should give more possibilities for finding music that actually feels connected to the story.

Music and emotion

Music can create different emotions through its tempo, energy, melody and vocals. Because of this, it can add another layer to the reading experience and support the atmosphere that the story is already creating.

Research about emotionally matching music and text has shown that readers can feel more immersed when the mood of the music fits the mood of the story. This supports the main idea of my project: the music should not simply play in the background, but should match and support the feeling of the book.

Why the correct match matters

The match between the book and the music is important because the wrong type of song can easily break the atmosphere. For example, a very cheerful song during an emotional or tragic part of a book may feel completely out of place.

When the music has a similar mood to the story, it can support the reading experience instead. Some books already include suggested playlists, but these are often quite short compared to the time needed to finish the book. My project builds on this idea by creating a longer selection of music that still matches the atmosphere of the story.

Music can also be distracting

At the same time, music does not automatically improve reading for everyone. Some people prefer complete silence when reading, while others, including me, enjoy having music in the background.

Research has shown that music with lyrics can be more distracting during reading, especially when the lyrics are in the same language as the text. It has also been found that people who already have a habit of listening to music while reading can be less affected by this distraction.

A larger review of 65 studies also found that background sound can have a small negative effect on reading performance, with lyrical music and understandable speech causing some of the strongest distractions.

Because of this, my project is not meant to suggest that everyone should listen to music while reading. The target users are specifically readers who already enjoy this combination or would like to experiment with it.

Mood is subjective

Another challenge within this domain is that mood is not completely objective. Two people can read the same book and describe its atmosphere differently. One reader may see a story mainly as romantic, while another person might focus more on its sadness or tension.

The same happens with music. A song that feels relaxing to one person may feel emotional or even sad to someone else. Personal experiences, music preferences and memories connected to a song can influence how someone understands it.

This is one of the reasons why involving different people in the annotation of my dataset is important. Instead of deciding all of the moods completely by myself, I want several readers to give their opinion about the books. This should give me a broader understanding of how the moods are perceived.

It also supports my decision to work with multiple moods instead of forcing every book into one category. If a book can be described as romantic, melancholic and calm at the same time, keeping all of these moods gives a much better picture of its atmosphere than simply calling it a romance.

Connection to my project

Understanding this domain helps define what the project is actually trying to achieve. I am not only trying to recommend songs based on the genre of a book. Two books from the same genre can have completely different atmospheres. Instead, I want to focus on the emotional feeling behind the story and use that as the connection between the book and the music.

The final experience should therefore feel more like creating a soundtrack for the atmosphere of the book rather than simply creating a general music recommendation. If the mood of the book can be described well and the music can be matched to those moods, the

playlist could become another part of the reading experience and help the reader stay connected to the world and feeling of the story.

Sources

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Model Requirements

The main requirement of the model is to understand the mood of a book based on its description. Because one book can have several different feelings at the same time, the model should not be limited to predicting only one mood. Instead, it should be able to recognise multiple moods that together describe the atmosphere of the book.

The predicted moods will later be used as the connection between the book and the music. Because of this, the quality of the mood prediction is more important than simply assigning the book to a general genre.

Analytic Approach

The main task of the model will be **multilabel text classification**. The input will be the description of a book and the output will be a combination of mood labels such as *romantic, dark, calm, melancholic, mysterious* or *tense*.

I want to experiment with different ways of representing the book descriptions. One approach will be **TF-IDF**, which focuses on the importance of words inside the text. Another approach will be **Sentence Embeddings**, which should capture more of the meaning and context behind the complete description.

For the classification itself, I am planning to experiment with **Linear SVM** and **Logistic Regression**. By comparing several combinations, I can see which approach works better with my dataset instead of deciding on one model before testing it.

I also want the moods to have different strengths where possible. For example, a book could be strongly romantic, slightly melancholic and also calm. This could later help with the music matching because it gives more information than simply saying that all three moods are equally important.

Model Input and Output

The main input of the model will be a **book description**. Information such as the title and author can be stored in the dataset, but the description will be the main information used for predicting the mood.

The output will contain several mood labels connected to the book. These predicted moods will then be passed to the music matching part of the system, where songs with similar characteristics can be found.

For a new book, the process should therefore look like:
Book description → Mood prediction → Mood combination → Music matching → Playlist

Data Requirements

For the model to learn the connection between descriptions and moods, I need a dataset containing books together with mood annotations. I am planning to create part of this dataset myself with the help of readers.

Around 10 people can annotate approximately 5 books each, which would give me an initial dataset of around 50 books. I will prepare a list of mood labels beforehand so that everyone works with the same set of moods. The participants can select multiple labels for each book instead of choosing only one.

The dataset should at least contain:

- Book title
- Author
- Book description
- Selected mood labels

The descriptions should also have enough information to understand the atmosphere of the book. Very short or unclear descriptions may make the prediction less reliable.

For the music side, I will need information that can help describe the feeling of a song. This could include characteristics such as energy, tempo or other audio features. I am currently considering using **Librosa** to extract numerical information from audio that can later be connected to the book moods.

Model Evaluation

Since this is a multilabel problem, I should not only look at whether one exact label was predicted correctly. The evaluation also needs to check how many of the relevant moods were found and how many incorrect moods were added.

I can use measures such as **precision, recall and F1-score** to compare the different modelling approaches. I also want to look at individual predictions manually because mood is subjective and a prediction can still make sense even when it does not exactly match every original annotation.

The final model does not need to predict the mood perfectly, but it should produce a combination of moods that is reasonable enough to create relevant music recommendations.

Limitations and Requirements

One of the main challenges is the size of the dataset. Around 50 books is relatively small for a machine learning project, so I need to keep the first model realistic and experiment with approaches that can work with smaller datasets.

Another challenge is that mood is subjective. Different people may describe the same book differently. Instead of seeing this only as a problem, I want to use several annotators where possible so that the dataset represents more than only my own opinion.

The music data will also depend on what information can be accessed from the chosen music platform. If the required audio information is not directly available, I may need to change the way the songs are analysed or matched.

Overall, the model should be able to take an unseen book description, recognise several moods that describe its atmosphere and provide useful information for the music matching part of the project.